HASAN IQBAL

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Education

Seneca Polytechnic — Markham, Canada
Bachelor of Science in Computer Science — 09/2025 - Present

Sheridan College — Brampton, Canada
Advanced Diploma in Computer Engineering Technology — 09/2020 - 08/2024

Technical Skills

- **Programming Languages:** Python, C++, C, CUDA, JavaScript/TypeScript, SQL, Bash
- **AI / Machine Learning:** PyTorch, TensorRT, LLMs, Generative AI, NLP, Computer Vision, Model Training, Fine-Tuning, Model Serving, Distributed Training, MLOps
- **Software Engineering:** Data Structures & Algorithms, System Design, REST APIs, Microservices, Object-Oriented Programming, Design Patterns, Unit Testing, Git, CI/CD
- **Backend & Infrastructure:** Node.js, Express, Flask, FastAPI, PostgreSQL, Docker, Kubernetes, AWS, GCP, Distributed Systems, HPC, Parallel Computing

Experience

Cerebras Systems — Sunnyvale, California
Artificial Intelligence Engineer — 05/2026 - 08/2026

- Led the end-to-end production bring-up of the Upstage Solar LLM in just two weeks
- Automated deployment pipelines for large-scale LLM usage and created monitoring tools for inference latency
- Optimized PyTorch compilation to reduce model generation time, and added custom feature caching
- Optimized wafer allocation for GPT-OSS model serving from 12 wafers down to 5, cutting infrastructure costs by over 50% while maintaining performance
- Resolved distributed execution, networking, and model serving problems in production GPU clusters
- Developed EDSL, a custom CUDA-like language, including its core runtime and software stack

ACI Brands — Oakville, Canada
Database Administrator — 04/2023 - 08/2023

- Efficiently stored and retrieved data in managed database systems
- Worked with teams to design and implement database solutions, optimizing data access
- Implemented database performance monitoring and fine-tuned query performance
- Analyzed retail data to ensure accurate reporting and identify trends

Canada Revenue Agency — Toronto West, Canada
Infrastructure Analyst — 09/2022 - 12/2022

- Evaluated systems and infrastructure for compliance with government policies and regulations
- Prepared documentation for system setups and operating procedures
- Assisted staff with infrastructure-related inquiries and issues

Ministry of Government and Consumer Services — Toronto, Canada
Executive IT Support — 01/2022 - 04/2022

- Identified and applied strategies to enhance executive functioning efficiency
- Led strategic initiatives to improve consumer service delivery and efficiency
- Worked closely with other departments to provide policy recommendations on government services

Ripple — Toronto, Canada
XRPL Campus Ambassador — 02/2023 - 02/2025

- Assessed blockchain security threats by conducting extensive analysis of untested code and unprotected keys to reduce risk
- Worked with the marketing team to create promotional materials for campus activities
- Implemented smart contracts written in Solidity for decentralized applications

Sheridan Student Union — Brampton, Canada
Student Board of Director — 04/2021 - 04/2022

- Managed and coordinated the Board's agenda and schedule, tracking performance and progress
- Suggested ways to improve proposed plans and make them more effective
- Made improvements to resolve issues and led key operational updates

Projects

Physics-Informed Neural Network (PINN) Humanoid Robot — 05/2026 - Present
github.com/redhasanh1/pinn-humanoid

- Running the Physical AI software stack for a 16-week capstone project to create a full-size autonomous humanoid robot, 3D print it, and deploy it in the real world
- Improved data efficiency for sim-to-real transfer using physics-informed neural networks that integrate data and physics-based loss terms for trajectory prediction and plan validation
- Developed a real-time ball-catching controller based on predicting object trajectories with physics-informed motion models
- Implemented ESP32 microcontroller-based finger control and teleoperation of the robot's 3D-printed hands, and trained an imitation-learning policy for autonomous task execution

ObjectRemoverAI — 10/2025 - 01/2026
ObjectRemoverAI

- Shipped a native iOS app in Swift for review to the App Store, managing account creation, authentication, and Apple review compliance
- Trained a custom YOLOv8 detector and integrated SAM2 for temporal mask tracking, feeding a ProPainter/RAFT inpainting pipeline for frame-consistent object removal
- Wrote a custom DCNv4 CUDA kernel (TensorRT plugin for Ada Lovelace SM 8.9), reducing inference latency 3x to 7ms per frame versus the PyTorch baseline
- Scaled the backend across a distributed cluster of workers using Celery and Redis, scaling linearly (tested on Salad.com cloud hardware, RTX 3060 through RTX 5090)

RoomFinderAI — 05/2025 - 08/2025
roomfinderai.com

- Developed a Node.js 18/Express 5 full-stack rental website with Supabase (PostgreSQL, Auth, Storage, Realtime) as the backend data layer
- Built an AI negotiation engine that detects rate limits on the OpenAI provider and automatically switches to Groq, keeping the negotiator live through provider outages
- Integrated Stripe Checkout for subscription billing and Google OAuth for sign-in, plus native companion Android (Java) and iOS (SwiftUI) apps
- Deployed on Railway using a Cloudflare Worker for edge routing and Cloudflare Turnstile for bot protection

Ripple Smart Lock Project — 01/2022 - 04/2022
github.com/StephanVorster/DecentLockFirmware

- Developed and 3D fabricated a physical housing for a high-security smart lock, incorporating XRPL blockchain authentication and biometric fingerprint verification
- Wrote Arduino Nano firmware in C/C++ to operate the lock mechanism independently of the main Raspberry Pi unit, keeping the lock operational if the Raspberry Pi or network fails
- Built cloud-backed access logging on AWS S3, a companion app in Kotlin/Java, and a Qt-based desktop app for lock configuration
- Implemented and tested XRPL XLS-30d NFTs for unique, traceable digital keys, deployed and demonstrated in Ripple's offices